



Nb₂CT_x MXene — Multilayer Powder

Research-grade two-dimensional transition-metal carbide powder

Nb₂CT_x MXene is supplied as a multilayer powder derived from Nb₂AlC by HF etching. The material has a nominal flake / particle size of 2–20 μm, typical conductivity of 1–10 S/cm, and mixed –O, –OH and –F surface terminations. This grade is supplied for electrochemical, electrocatalysis, conductive-composite and interface research.

KEY FEATURES

- Layered multilayer powder derived from Nb₂AlC
- Nominal size: 2–20 μm
- Typical conductivity of 1–10 S/cm
- Mixed –O, –OH, –F surface terminations

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal–sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

QUALITY / DOCUMENTATION

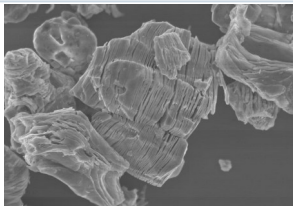
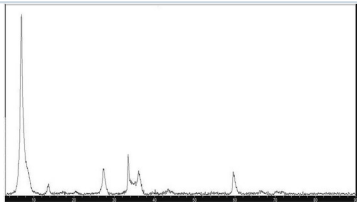
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	Nb ₂ CT _x MXene
Precursor	Nb ₂ AlC
Product form	Multilayer powder
Preparation	HF etching
Layer structure / count	Multilayer
Nominal flake / particle size	2–20 μm
Typical conductivity	1–10 S/cm
Surface terminations	–O, –OH, –F
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION

	Multilayer Nb ₂ CT _x — SEM. Representative characterization; not lot-specific.
	Multilayer Nb ₂ CT _x — XRD. Representative characterization; not lot-specific. Non-data annotation removed for presentation; diffraction trace and axes otherwise unchanged.

Product positioning: Supplied and technically supported by WiniGen Materials for research and development use.